

A Comprehensive Comparative Study of Optimization Algorithms on Heart Disease Prediction

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Introduction

Heart disease is one of the leading causes of global mortality, making early prediction and medical risk assessment extremely important. Machine learning models, when optimized correctly, can significantly improve the accuracy and reliability of diagnosis. This project applies a wide range of optimization techniques—both constrained and unconstrained—to the Heart Disease UCI dataset, a standard benchmark dataset for cardiovascular risk prediction.

Dataset Description

We use the **Heart Disease UCI Dataset** from Kaggle, containing patient medical records with the following clinical features:

- **age**
- **sex**
- **cp** (chest pain type)
- **trestbps** (resting blood pressure)
- **chol** (serum cholesterol)
- **fbs** (fasting blood sugar)
- **restecg** (resting ECG results)
- **thalach** (maximum heart rate achieved)
- **exang** (exercise-induced angina)
- **oldpeak** (ST depression induced by exercise)

These features collectively help predict the presence of heart disease.

Dataset link: <https://www.kaggle.com/code/mragpavank/heart-disease-uci/notebook>

Project Objective

The goal of this project is to implement, analyze, and compare multiple optimization approaches for training logistic regression models and visualizing the optimization process. Special emphasis is placed on understanding how different algorithms behave, converge, and satisfy optimality conditions.

Techniques Implemented

Unconstrained Optimization

- Gradient Descent
- Stochastic Gradient Descent
- Mini Batch GD
- Momentum
- Nesterov Accelerated GD
- Armijo
- Exact Line Search
- Adagrad
- RMSProp
- Adam
- Newton's Method
- Quasi-Newton (BFGS and L-BFGS)

Constrained Optimization

- Equality constraints
- Inequality constraints
- Mixed (equality + inequality) constraints
- KKT (Karush-Kuhn-Tucker) optimality conditions
- LICQ (Linear Independence Constraint Qualification) verification

Visualization

To provide deeper insight into the optimization dynamics:

- Cost function contours and 3D surfaces were generated
- Optimization trajectories were plotted
- Feasible regions were shaded for inequality constraints

Key Outcomes

- Successfully trained logistic regression models under different optimization regimes
- Observed convergence patterns for multiple algorithms
- Verified KKT conditions for constrained optimization problems
- Demonstrated how constraints affect the feasible region and optimal solution
- Identified which optimization methods perform efficiently for medical prediction tasks
- Validated accuracy of models on test data

Conclusion

This project provides a complete end-to-end exploration of optimization in machine learning using the Heart Disease UCI dataset. By combining multiple optimization techniques, mathematical constraint handling, and powerful visualizations, we gain a deeper understanding of how models learn, converge, and adhere to real-world constraints. The project highlights the importance of optimization theory in building reliable and interpretable AI models, especially in sensitive applications like healthcare.